CHEMISTRY | ADAPTLY

RATES OF REACTION – COLLISION THEORY

KEY VOCABULARY

Rate of reaction: how quickly reactants are used up or products are made.

Activation energy: the minimum energy needed for a successful collision.

Catalyst: speeds up a reaction without being used up.

Collision theory: particles must collide with enough energy to react.

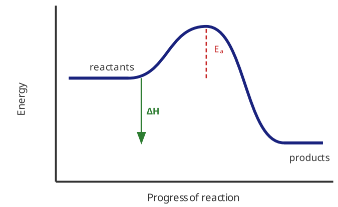
COMMON MISCONCEPTIONS

~~A catalyst is used up in the reaction.~~
✓ A catalyst is unchanged and can be reused.

~~Heating gives particles more particles.~~
✓ Heating gives particles more energy, not more particles.

~~A catalyst increases activation energy.~~
✓ A catalyst lowers the activation energy.

REACTION PROFILE (EXOTHERMIC)



Products are lower than reactants, so energy is **released**. A catalyst lowers E_a.

1 COLLISION THEORY

State the **two conditions** needed for a successful collision. **[2 marks]**

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2 EFFECT OF TEMPERATURE

Explain, using collision theory, why increasing temperature increases the rate. **[3 marks]**

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3 EFFECT OF CONCENTRATION

Explain why a higher concentration of acid speeds up a reaction. **[2 marks]**

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4 MEASURING RATE

48 cm^3 of gas is collected in 24 s. **Calculate** the mean rate of reaction. **[2 marks]**

Answer: ______________________________ cm^3/s

5 CATALYSTS

Complete the sentence using the word bank. **[2 marks]**

A catalyst provides a different pathway that _____ the activation energy, making the reaction _____.

lowers | faster | raises | slower

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6 INTERPRET A GRAPH

On a volume–time graph, two curves level off at the same height but one is steeper. **Explain** what this shows. **[2 marks]**

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TIP: *More frequent collisions with enough energy = a faster rate.*

✓ **Think: Why does the rate slow down as a reaction proceeds?**